Introduction to Databricks

A unified platform for modern data engineering, analytics, and machine learning

Agenda

01

Why Databricks

Understanding the modern data challenges driving platform evolution

02

Architecture Evolution

From siloed systems to unified lakehouse architecture

03

Problem Solutions

How Databricks addresses enterprise data pain points

Modern Data Challenges

Data Explosion

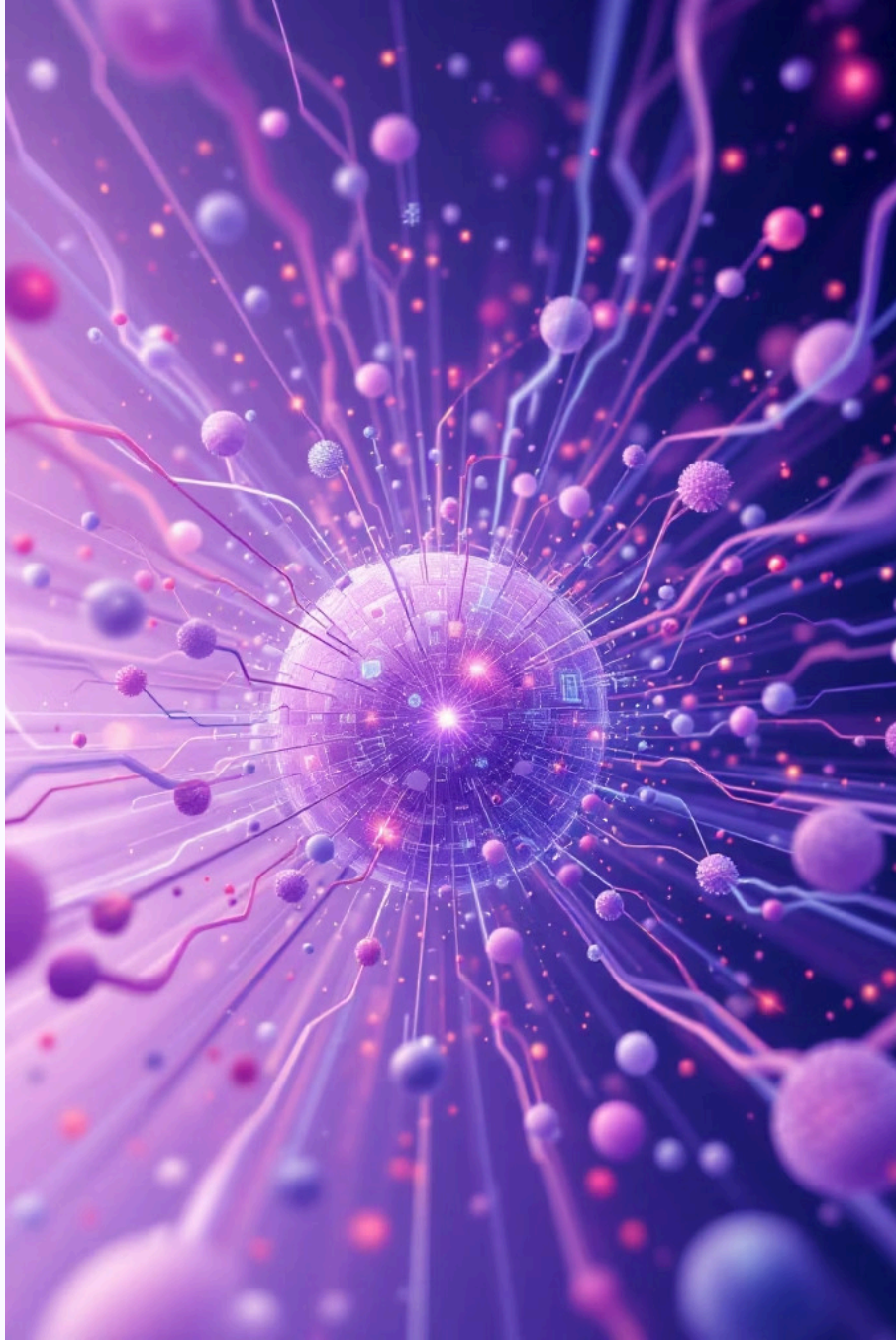
Exponential growth in volume, velocity, and variety of data requiring scalable infrastructure

Multiple Systems

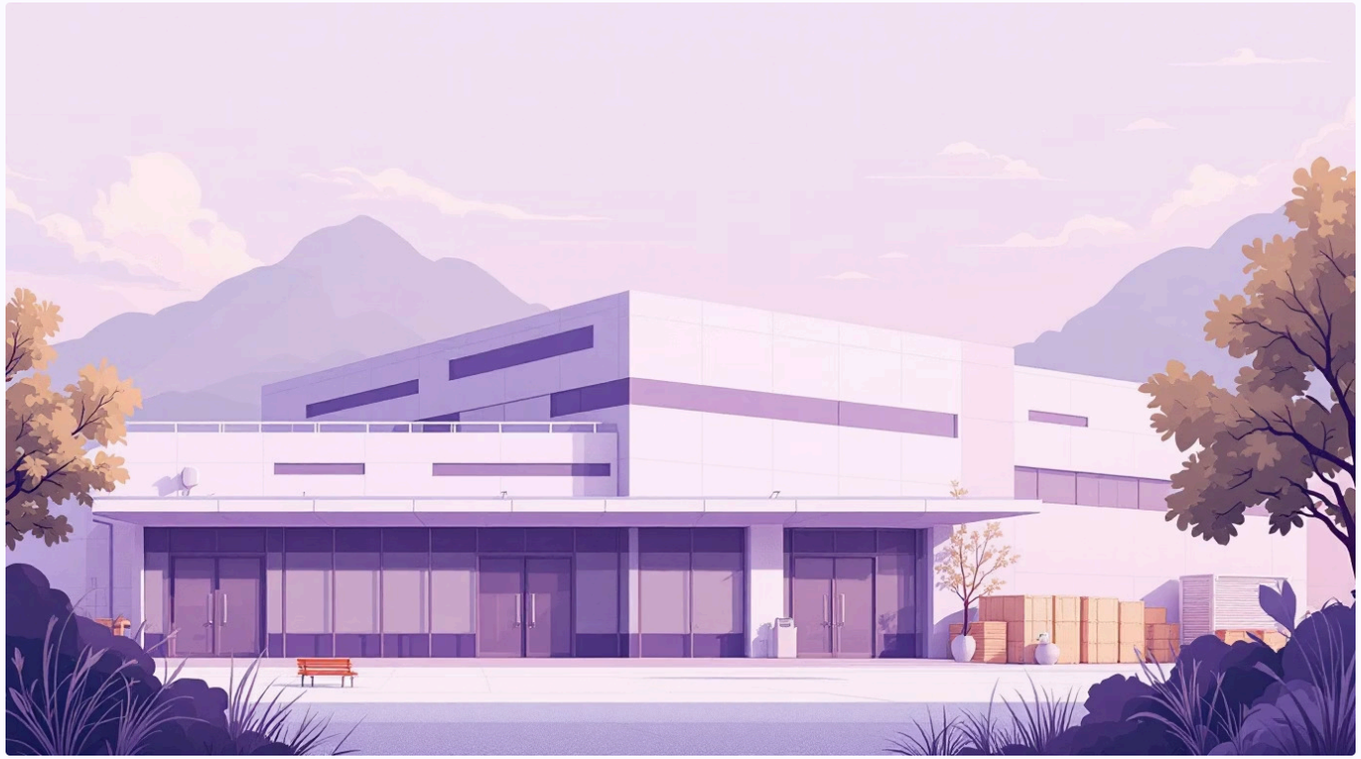
Disconnected data warehouses, lakes, and streaming platforms creating complexity

Latency & Cost

Slow query performance and expensive infrastructure limiting business agility



Data Warehouse Limitations



Expensive Infrastructure

High costs for storage, compute, and licensing that scale poorly with data growth

Not ML-Friendly

Limited support for machine learning workflows and unstructured data processing

Rigid Schema

Inflexible structure requiring predefined schemas that slow down development

Data Lake Limitations

Absence of ACID Transactions

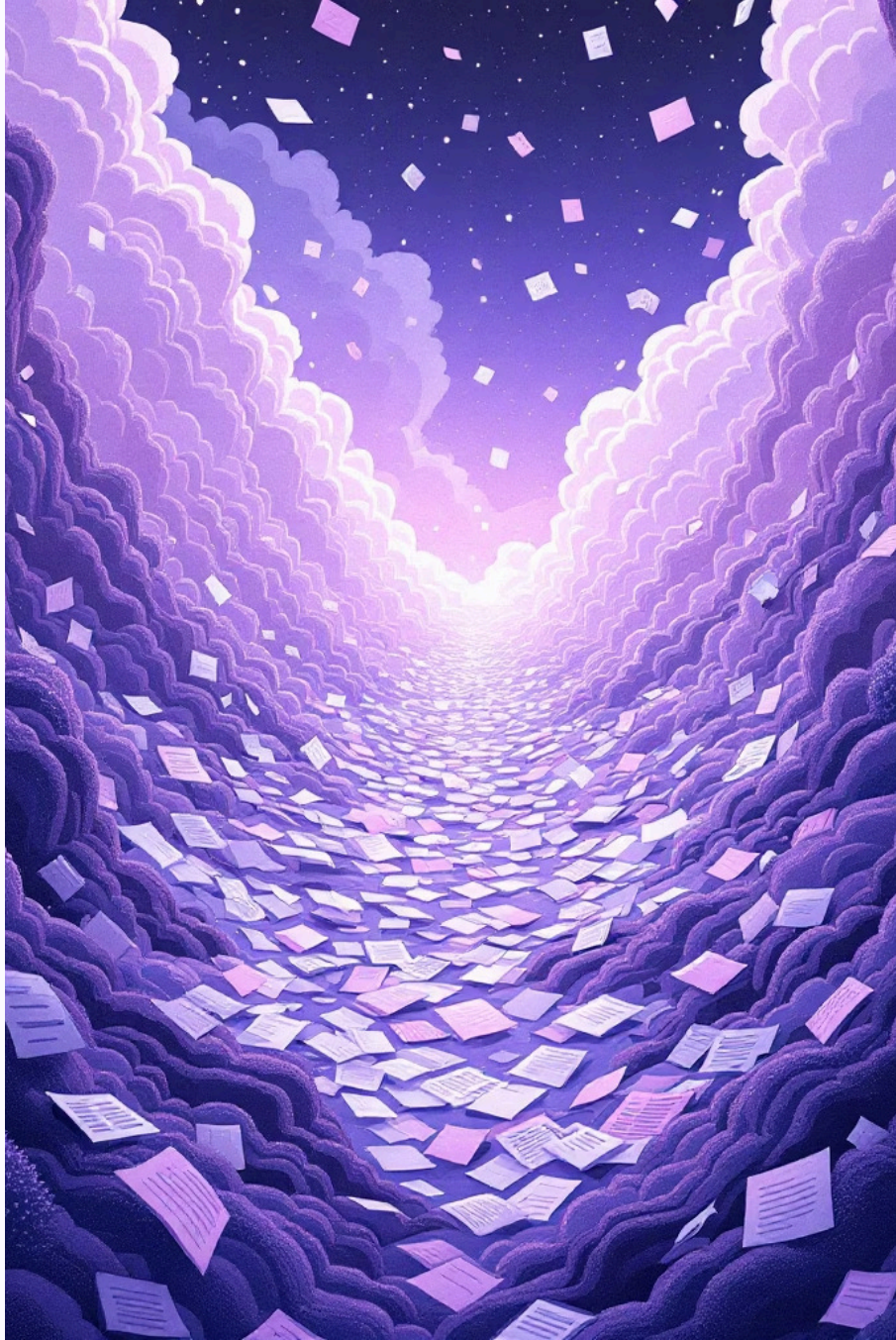
No transactional integrity guarantees, leading to data corruption risks and inconsistent reads

Poor Query Performance

Slow SQL queries and analytics due to lack of indexing and optimization layers

Data Chaos

No governance, metadata management, or data quality controls creating unreliable datasets



The Big Problem

Two Systems = Duplication + Complexity

Organisations forced to maintain separate data warehouses and lakes leads to:

- Data duplication across multiple storage systems
- Complex ETL pipelines and synchronization challenges
- Higher costs for infrastructure and maintenance
- Slower time-to-insight for business users

Evolution of Architecture

1

Data Warehouse

Structured data, SQL queries, expensive infrastructure

2

Data Lake

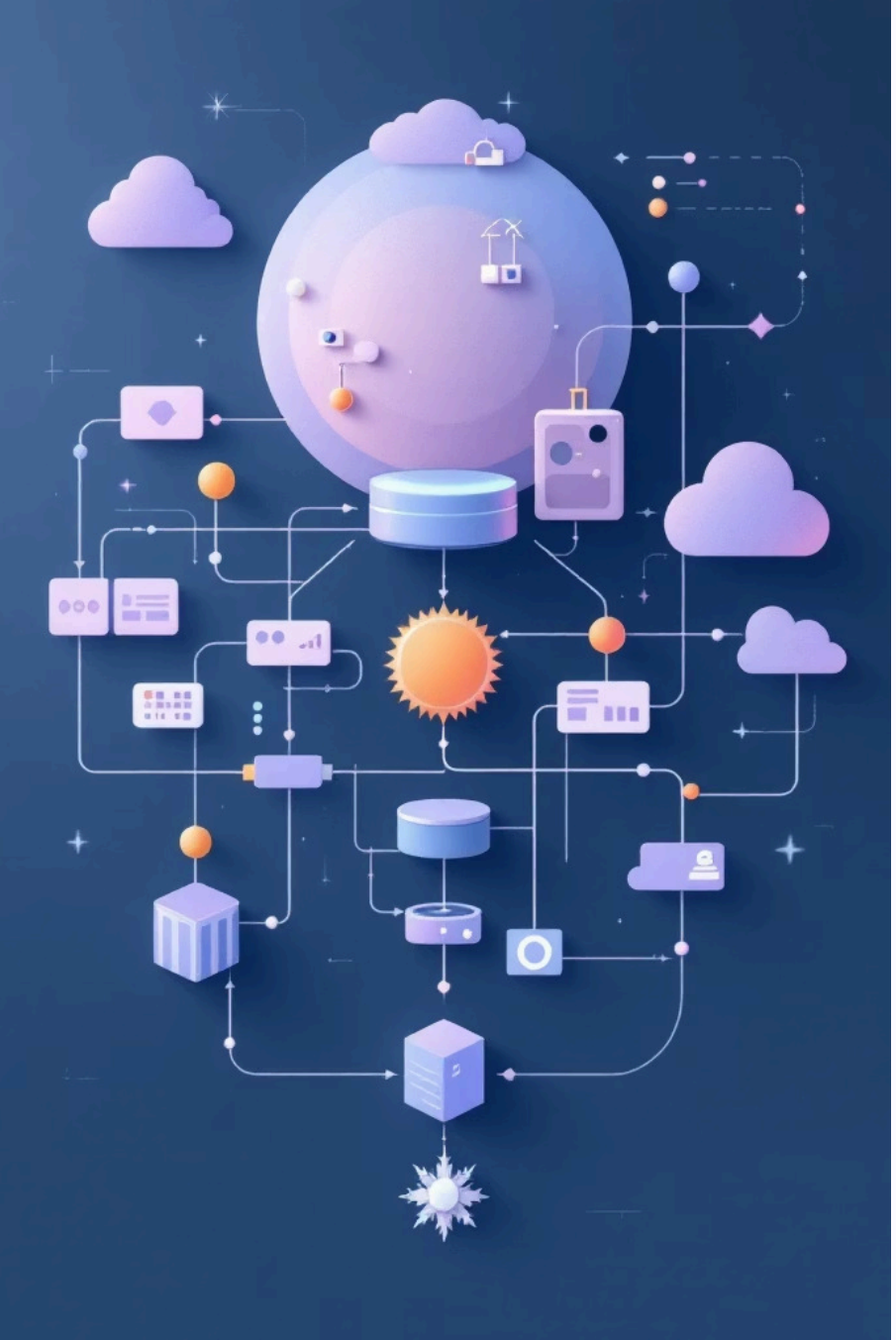
Raw data storage, schema-on-read, no governance

3

Lakehouse

Unified platform, ACID transactions, open formats

Each evolution addressed previous limitations whilst introducing new capabilities for modern data workloads



Why Databricks Exists

Simplify Architecture

Replace multiple systems with single platform reducing complexity and operational overhead

Unify Workloads

Support analytics, data science, and machine learning on common infrastructure and tools

Reduce Costs

Leverage cloud economics with scalable compute and cost-effective storage options

What Databricks Provides

Unified Platform

Single environment for data engineering, analytics, and machine learning with collaborative workspaces

Spark + ML + SQL

- Apache Spark for distributed processing
- MLflow for machine learning lifecycle
- SQL interface for analytics and BI
- Delta Lake for reliable data storage

Databricks: Our Journey of Innovation

1

2013: Founding & Spark

Founded by the creators of Apache Spark from UC Berkeley AMPLab, bringing distributed processing to the forefront.

2

Early Funding Rounds

Rapid early growth and significant series funding propelled expansion and platform development.

3

2023: MosaicML Acquisition

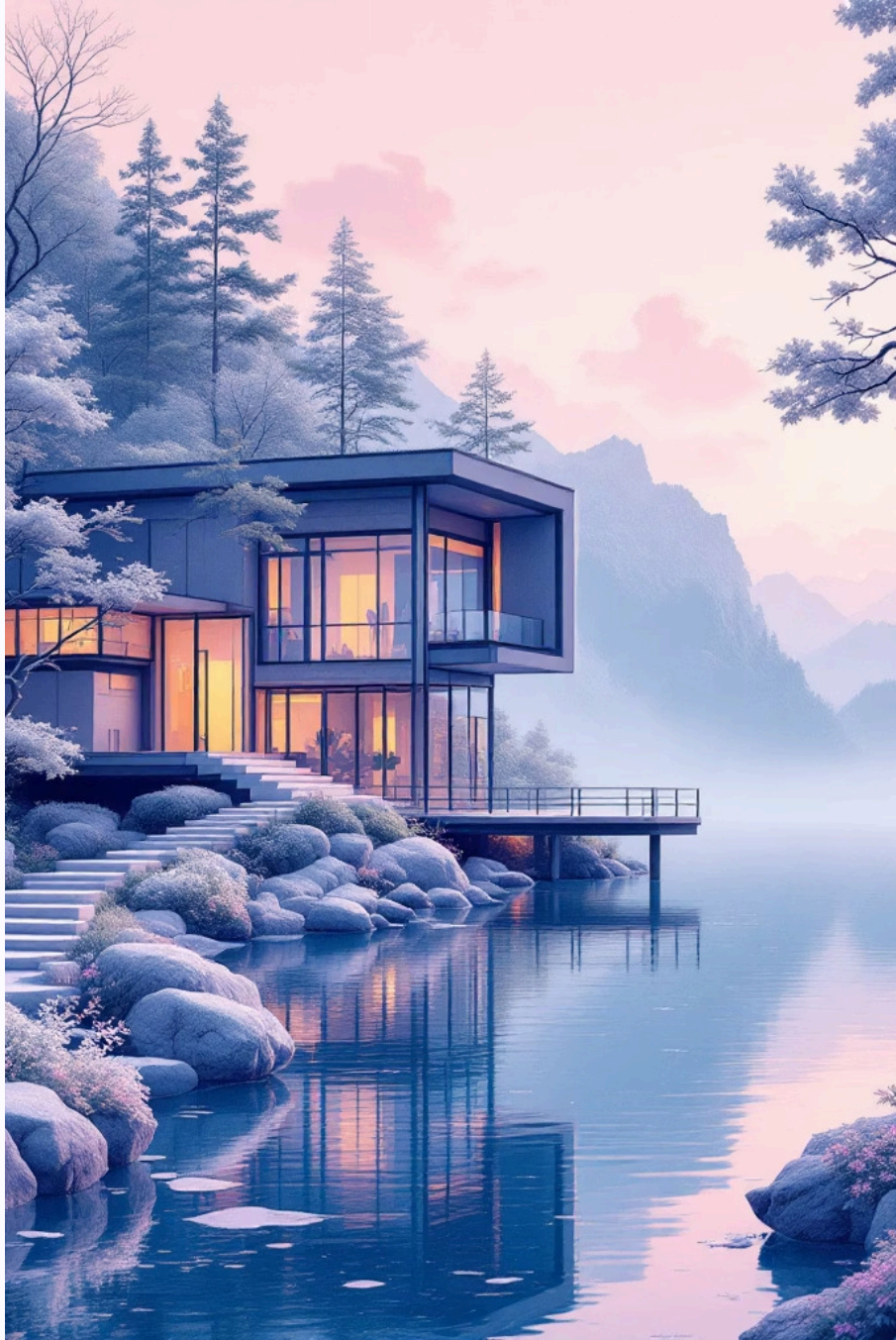
Acquired MosaicML to integrate generative AI capabilities, accelerating innovation in AI solutions.

4

Continued Growth & Lakehouse Leadership

Evolved into a leading data and AI platform, championing the open Lakehouse architecture for unified data management.

Understanding Lakehouse

Lakehouse represents the core architectural concept behind Databricks, combining the best of data lakes and warehouses

> **Next:** Deep dive into Lakehouse architecture and how it solves enterprise data challenges